



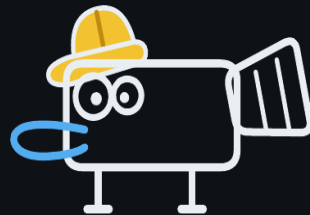
MURDER BOARD · ROUND 1 · FOR PRATEEK

Your questions, answered. One question per slide.

What you asked, how I answer it, and the numbers that justify it.

proto plans · Baker makes · Mixer tests · Probe Matrix reaches every net

Soham, founder · 2 Oct 2026 · protopus-kappa.vercel.app



BEFORE WE START

Sir, your feedback changed the pitch. Here is what is different now.

PROTO FIRST

Proto is the company. [Baker](#), [Mixer](#) and the [Probe Matrix](#) are its hands and eyes.

PROTOTYPING, NOT PRODUCTION

The first 1–10 boards of a design. Volume still goes to a fab.

SOFTER CLAIMS

Lower cost per try, shorter lead time. Every number is marked projected.

BAKER FIRST

₹15 lakh for nine months, or ₹10 lakh lean. [Mixer](#) waits for step 2.

PAYBACK

47 boards against JLCPCB: about six months for a startup.

HONEST SCALE

750–1,250 boards a year per [Baker](#). Beyond that, a fab is the right tool.

QUESTION 01 · PROTO IS THE MOAT

SIR, YOU ASKED

“The strongest moat is hidden in the idea. If I were the founder, I would pitch Proto instead of two hardware robots. And don’t project it as a desktop PCB mill; that has a rough history.”

I CAN ANSWER

You’re right, sir: Proto is the company. But an AI can only build hardware if it has hands and eyes in the real world. Software AI reaches every tool through APIs. Hardware has no API, so [Baker](#) and [Mixer](#) are that API. And [Baker](#) isn’t a mill: a mill does one step for experts. [Baker](#) runs fifteen, with Proto guiding a first-timer.

TO JUSTIFY



AI today: Flux gives Gerbers,
Claude drives Fusion

protopus: the AI's hands and eyes

THE HANDS

[Baker](#)

makes the board: mill,
mask, stencil, paste,
reflow

THE EYES

[Mixer](#)

signal generator +
analysers; replays
real sensor faults

THE FINGERS

[Probe Matrix](#)

probes every net, so
Proto can test and
program

THE MOAT IS THE DATA every run logs design → machine settings → measurement → outcome. A file-only AI never sees the real board, so it can never collect this.

NOT A DESKTOP MILL A mill automates one step for people who are already experts. Bantam Tools stopped selling its PCB mill in 2025. [Baker](#) runs 15 steps in one frame, from bare FR4 to a reflowed board, and Proto plans each one.

QUESTION 01 · PROTO IS THE MOAT

SIR, YOU ASKED (CONTINUED)

“Why is Proto the moat, and not the machines?”

I CAN ANSWER

Because ease is the market, sir. People learn hardware by doing: try, look, adjust. The faster that loop turns, the faster they learn. Today every rung of hardware knowledge loses people. Proto takes rungs away, and each rung we remove brings in a new crowd: students, designers, first-time founders.

7–20 days

per try today, by JLCPCB to India

≈2 hours

per try on Baker (target)

TO JUSTIFY

RUNG	WHERE PEOPLE GET STUCK	WHO CARRIES IT NOW
1 idea → spec	vague goals, scope creep	Proto asks, then writes the spec
2 parts	hours of datasheets; stock runs out	Proto picks in-stock parts
3 schematic	errors that only show on the bench	Proto + Flux check it
4 layout	weeks to learn an EDA tool	Proto fixes it (Baker rules)
5 fabrication	7–20 days to India per try	Baker, in hours
6 assembly	SMD soldering, stencils, reflow	Baker + a printed jig
7 bring-up	you can't tell what's wrong	Mixer + Probe Matrix (step 2)
8 iterate	slow loops, fewer attempts	Proto explains the fix

Rungs 1–4 are already being solved in software. Rungs 5–8 are physical, which is why AI-only tools stop at the file. Lovable reached \$100M ARR in 8 months by removing the coding rung.

QUESTION 02 · SOFTER CLAIMS

SIR, YOU ASKED

“Avoid strong claims like “replaces a fabrication order”. An investor who has invested in hardware manufacturing: how would he take it? Softer claims are better, like lower cost and faster lead time.”

I CAN ANSWER

Fair point, sir, and we have changed it. We never meant to replace fabs. We fill the gap between a finished design and a board you can actually test. It's like 3D-printing a bottle before you pay for the injection mould: [Baker](#) makes the first few boards, and the fab still makes the thousands. It also gets files that have already worked on a real board.

TO JUSTIFY

WE USED TO SAY	NOW WE SAY
“replaces a fabrication order”	lower cost per try and same-day lead time for the first 1-10 boards
“send it at night, pick it up in the morning”	about 2 hours per board, 15-20 minutes hands-on (target)
“fully automatic”	steps 1-12 automatic; you place parts and start the reflow
“replaces your bench”	Analog Discovery 3-class test, run by Proto (step 2)

THE PATH FROM IDEA TO 1,000s



Every motor speed, processing time and jig-print time is a projected engineering estimate. The proof of concept measures them.

QUESTION 03 · ONE PRODUCT AT A TIME

SIR, YOU ASKED

“Two products for a first-shot raise of ₹10L is a bit aggressive. We should go step by step. Maybe start with Baker first?”

I CAN ANSWER

Agreed, sir. We go one product at a time, starting with Baker and Proto. The two machines stay in the vision: one makes the hardware, the other tests it, programs it and feeds back to Proto. But the money we raise now builds only Baker. Mixer and the Probe Matrix come in step 2, after the proof of concept.

TO JUSTIFY

STEP 1 · NOW · ₹15L

Proto + Baker

20 boards, cycle time measured

STEP 2 · SEED

Mixer + Probe Matrix

~₹1.5 Cr seed; pilots built turnkey in China

STEP 3 · ECOSYSTEM

Proto runs it all

plus any 3D printer

WHERE ₹15,00,000 GOES · 9 MONTHS · NO FOUNDER SALARY



IF THE ROUND IS ₹10L Baker ₹3.73L + Rev B ₹1.0L + engineer for 6 months ₹3.0L + compute ₹0.9L + testing ₹1.0L + reserve ₹0.37L. No lab set-up, IP or outreach yet.

QUESTION 04 · COMPETITION AND MARKET

SIR, YOU ASKED

“I think a thorough competition analysis and market study is needed.”

I CAN ANSWER

We did the study, sir. Every competitor solves one step: a mill makes a bare board, a placer places parts, a USB instrument measures, a fab ships in weeks, and an AI tool stops at the file. Nobody runs the whole loop, and nobody lets an AI drive the machines. That is where we are different. Our beachhead alone is about 8,000 buyers in India.

TO JUSTIFY

₹6.4L cr

global PCB

₹3.1L cr

global test + measurement

₹55K cr

India PCB, ~15% a year

₹8.3K cr

India test + measurement

BEACHHEAD 5,868 AICTE colleges + ~2,250 hardware startups = 8,118 buyers → ₹243 Cr of **Baker** (₹271 Cr with **Mixer**) + ₹8.1 Cr a year of Proto plans. Then 10,000 Atal Tinkering Labs (+50,000 planned).

	MAKE	ASSEMBLE	TEST	GUIDE
Desktop mills (Carvera, LPKF)	●	○	○	○
Ink printer (Voltera)	●	●	○	○
Pick-and-place (Neoden)	○	●	○	○
USB instruments (AD3, Moku)	○	○	●	○
Fabs (JLCPCB, PCBWay)	●	●	○	○
AI design (Flux)	○	○	○	●
protopus: proto + baker	●	●	●	●

QUESTION 05 · AUTOMATION, HONESTLY

SIR, YOU ASKED

“Send the design at night, pick it up in the morning’ is a bit too aggressive. We have to load boards, place components, etc. So it’s not fully automated.”

I CAN ANSWER

You’re right, sir, it isn’t fully automatic, and we no longer say it is. Boards don’t need loading every time, because a refill magazine feeds the next blank. The placement jig is generated from the Gerber and printed in parallel. You drop the parts into the jig and set it on the reflow plate. Start a board after lunch, and test it the same afternoon.

TO JUSTIFY

STEP	WHAT HAPPENS	WHO	MIN
01-02	feed from the refill magazine, probe the height map	auto	5
03	isolation mill + drill	auto	30-55
04-09	clean, coat the mask, UV image, develop, cure	auto	16-27
10-12	laser-cut the stencil, paste, release	auto	8-13
jig	generated from the Gerber, printed in parallel	auto	+0
13	you drop ~40 parts into the jig	you	7-14
14	you set the jig on the plate; reflow runs itself	you + auto	10-12
15	test: Proto runs the plan	you hook up	5-15

81-141 min	15-20 min	3-5
per board (target)	of it hands-on	boards per 8-h shift

QUESTION 06 · COMPETITORS' COSTING

SIR, YOU ASKED

"I also strongly recommend looking at the competition and their costing."

I CAN ANSWER

Here is what the same loop costs elsewhere, sir. Bought as separate machines (a mill, a placer and a USB instrument) it comes to about ₹7.8 lakh before duty, and it still has no solder mask, no stencil and no AI tying it together. [Baker](#) with Proto is ₹2.99 lakh. Adding [Mixer](#) brings it to ₹3.34 lakh.

TO JUSTIFY

PRODUCT	LIST PRICE	WHERE IT LOSES TO US
LPKF ProtoMat	₹8.6-24.5L	separate machines for each step
Voltera V-One	~₹3.1L	ink traces; no mask or test
Carvera / Air	₹4.8L / ₹2.2L	mills only
Bantam Tools mill	discontinued	one step, for experts
Neoden YY1	~₹2.6L	places parts only
Digilent AD3	~₹34,000	a human reads the screen
JLCPCB → India	₹6,721 / try	7-20 days + customs
Flux.ai	₹1.3-4.3k / mo	stops at the file

THE SAME LOOP, BOUGHT SEPARATELY vs PROTOPUS

mill + placer + AD3		₹7.82L, no AI
Baker + Mixer		₹3.34L, the whole loop
Proto + Baker		₹2.99L, step 1

QUESTION 07 · WHAT IT COSTS, AND PAYBACK

SIR, YOU ASKED

“How much is this machine going to cost? Say X lacs: we also need to see how many boards are created and whether it cancels the cost.”

I CAN ANSWER

Baker will be ₹2,99,000 (projected). On Baker a board costs about ₹334 in materials. The same assembled board from JLCPCB, landed in India, costs ₹6,721 and takes 7–20 days. So Baker pays for itself after 47 boards. That’s about six months for a startup doing two boards a week, and a year for a college lab doing one.

Startup: 2 boards a week; lab: 1 a week. Engineer time and the 7–20 days of waiting are not counted, so the payback is conservative. JLCPCB: 5 PCBs + PCBA + DHL + 10% BCD + 18% IGST.

TO JUSTIFY

₹2,99,000	₹334	₹6,721			
Baker list (projected)	materials per board	JLCPCB, landed			
CUSTOMER	BUYS	VS	BOARDS	STARTUP	LAB
<u>Baker</u>		JLCPCB	47	5.9 mo	11.8 mo
<u>Baker</u>		Indian 24-h fab	58	7.3 mo	14.5 mo
<u>Baker</u> + <u>Mixer</u>		JLCPCB	53	6.6 mo	13.3 mo

IF THE PRICE MOVES · BOARDS TO PAY BACK BAKER

47	55	63	71
at ₹2.99L	at ₹3.49L	at ₹3.99L	at ₹4.49L

FROM OUR SIDE each Baker earns ₹1,54,158 of gross profit (51.6%), about ₹99,000 after all costs. 10 Bakers return the whole ₹15 lakh round on gross profit; 16 on all-in profit.

QUESTION 08 · SCALE

SIR, YOU ASKED

“Innovation is good but may not be scalable in a production environment. 1,000 boards? 10,000? 1 lakh?”

I CAN ANSWER

Baker isn’t meant for production, sir, and we won’t claim it is. It’s for the first one to ten boards of a design. At 10,000 or a lakh boards, a fab is the right tool, and Proto hands it files that have already worked. What scales is the number of people who can prototype, and our own manufacturing, which is turnkey in China.

TO JUSTIFY

BOARDS A YEAR	BAKERS	RIGHT TOOL
10	1	<u>Baker</u> : same-day loops beat fab minimums
100	1	<u>Baker</u> : one machine covers a busy lab
1,000	1-2	<u>Baker</u> + a fab for identical batches
10,000	14	fab / EMS, with Proto’s checked files
1,00,000	134	fab / EMS: not what <u>Baker</u> is for

HOW WE SCALE

China, turnkey

parts, assembly,
factory test

India

final test, minimal
assembly, ship

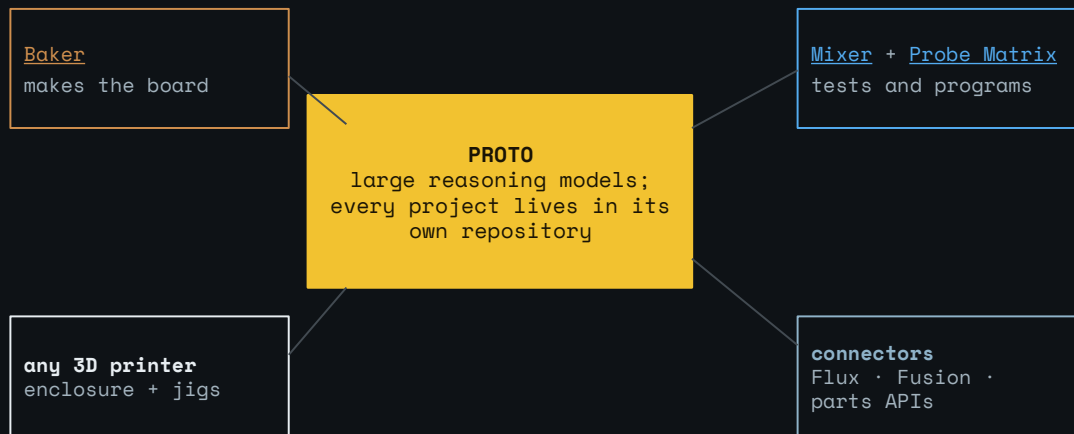
Proto

software: each extra
user costs only compute

One Baker: 750–1,250 boards a year (3–5 per shift, target). Production cost ₹1,44,842 per Baker at 100 units; RFQs planned at 5–5,000 units. India first, then global.

AND WHERE THIS IS GOING

Sir, the vision: one plain-English brief in, one tested prototype out.



We never sell a 3D printer: Proto drives the one you have. BOMs come with purchase links; the case is sized from the measured board.

The one-stop solution for prototyping: **idea to innovation, without the friction.**

EXAMPLE · A SMARTWATCH WITH 10 BUTTONS

- 01 you describe it in plain English
- 02 Proto checks feasibility and limits, writes the requirements
- 03 Proto designs the PCB, sensors and BOM with buy links
- 04 Baker makes the board; your printer prints the case
- 05 Proto flashes the firmware; Mixer + Probe Matrix test it and flag failures
- 06 you say what felt off; Proto plans the next revision



Thank you, sir.

Happy to get on a call and go through any of this in detail.

[website](#) · [Baker](#) · [Mixer](#) · [Probe Matrix](#)

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